Companion to Bird & de Moor, Algebra of Programming

§5. Datatypes in Allegories

§5.1. Relators

(5.1a)

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from R to S IS R⊑S. A **relator** F : $\mathcal{C}\to\mathcal{D}$ is a 2-functor between allegories:

$$F(\mathbb{1})=\mathbb{1} \qquad F(RS)=F(R)F(S) \qquad R \sqsubseteq S \implies F(R) \sqsubseteq F(S)$$

Preserving ° is **not** asked for — ° is an identity-on-objects involution $\mathcal{C}^{op}\to\mathcal{C}$, no part of the 2-category.

(5.1b)

	the statement
1	Map(f) ⟹ Map(F(f)) and Map(f) ⟹ F(f°) = F(f)° . Lemma 5.1
2	Over a **tabular** allegory a functor is a relator ⟺ it preserves °. Theorem 5.1 the book's reverse direction (preserving ° ⟹ relator) needs the source to be Rel: the counterexample over a tabular allegory that is not Rel shows it can fail
3	F(R°) = F(R)° for every R, so F(R)° needs no bracket.
4	Two relators agreeing on maps are equal. Corollary 5.1
5	Coreflexive(X) ∧ Coreflexive(Y) ⟹ F(X ∩ Y) = F(X) ∩ F(Y) Ex 5.2
6	F(R ∩ S) ⊑ F(R) ∩ F(S) , and strictly. Ex 5.2
7	PreservesRecip(F) ⟹ F(dom(R)) = dom(F(R)) Ex 5.5

The **power relator** P — xs P(R) ys⟺(∀a∈xs. ∃b∈ys. a R b)∧(∀b∈ys. ∃a∈xs. a R b) — is where the fourth is strict: for R={(a₁,b₁),(a₂,b₂)} and S={(a₁,b₂),(a₂,b₁)} the pair ({a₁,a₂},{b₁,b₂}) is in P(R)∩P(S), while R∩S=∅.

§5.2. Relational products

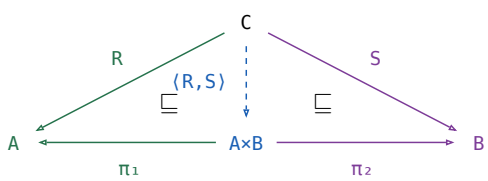
(5.1)

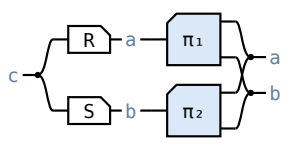
The **fork** of R : C→A and S : C→B is ⟨R,S⟩≜Rπ₁°∩Sπ₂° , where (π₁,π₂) is the tabulation of ⊤ .

(5.2a)

⟨R,S⟩ π₁ = dom(S) R (5.6) ⟨R,S⟩ π₂ = dom(R) S (5.7)

(5.2b)

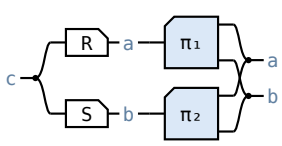




A domain is coreflexive, so ⟨R,S⟩π₁⊑R, with equality exactly when S is entire; for maps both triangles commute and ⟨f,g⟩ is unique. In Rel, c ⟨R,S⟩ (a,b) iff c R a and c S b — copy c, then R on one strand and S on the other, which is ◁(R⊗S) on the right.

No ° survives the translation. π₁=𝟙⊗⊸ discards the second component, so π₁°=𝟙⊗⊸ **creates** one out of nothing, and ∩ is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:

(5.2c)



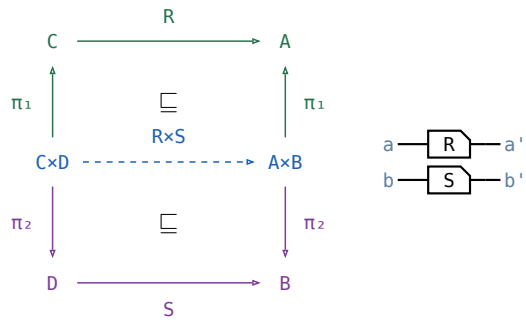
⊸▷=𝟙 on each half

§5.2.1. Relational product R×S

(5.2)

R×S≜⟨π₁R,π₂S⟩ , a relator in each argument but no longer a categorical product.

(5.2.1a)

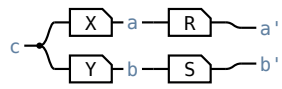


Right-then-up is `(R×S)π₁`, up-then-right is `π₁R`, and `(R×S)π₁⊑π₁R`, equality when `S` is entire. In `Rel`, `(c,d) (R×S) (a,b)` iff `c R a` and `d S b` — two strands side by side, no copy dot: `R×S=R⊗S`.

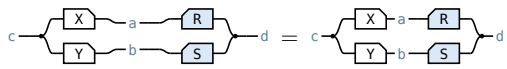
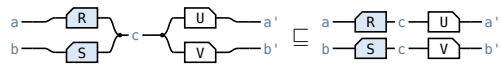
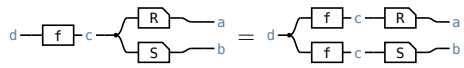
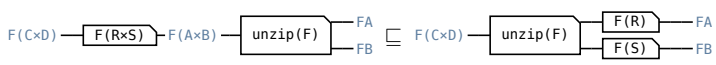
§5.2.2. Absorption

For `X : E→C` and `Y : E→D`, `⟨X,Y⟩(R×S)=⟨XR,YS⟩`. Both sides are this picture:

(5.3)

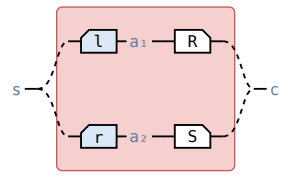
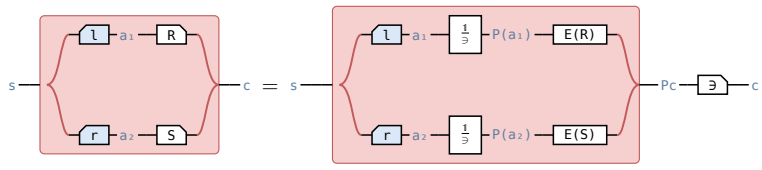
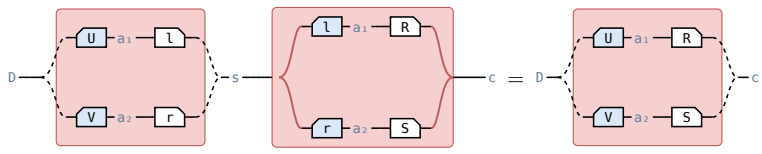


(5.2.2a)

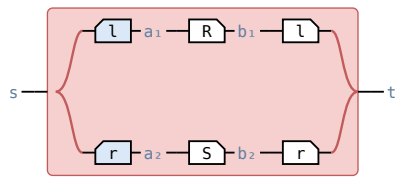
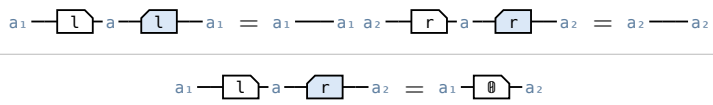
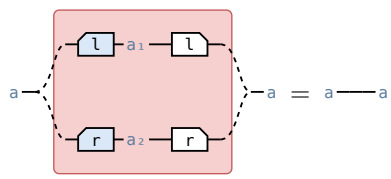
	the law	picture
1	`⟨X,Y⟩ ⟨R,S⟩° = (X R°) ∩ (Y S°)`	
2	`⟨R,S⟩° ⟨U,V⟩ ⊑ (R° U) × (S° V)`	
3	`Map(f) ⟹ f ⟨R,S⟩ = ⟨f R,f S⟩` `f` a map; it fails for an arbitrary arrow;	
4	`X ⟨R,S⟩ ⊑ ⟨X R,X S⟩` the half of the row above that holds for every `X`	
5	`F(R × S) unzip(F) ⊑ unzip(F) (F(R) × F(S))` `unzip(F)≜⟨F(π₁),F(π₂)⟩`; only `⊑` for an arbitrary `R`, `S`	
6	`g=curry(f)⟺(g×𝟙)eval=f` reading `×` as the relational product, does `Rel` have exponentials?	

§5.3. Relational coproducts

(5.3a)

	the statement R : a_1→c, S : a_2→c	**picture**
1	[R,S] ≜ (l° R) ∪ (r° S) (5.9) The tape is the union — a particle entering at A+B takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint.	
2	[R,S] = [$\frac{R}{∋}$,$\frac{S}{∋}$] ∋	
3	l [R,S] = R , r [R,S] = S l T = R ∧ r T = S ⟹ T = [R,S] ,	
4	U : a_1→D, V : a_2→D [U,V]° [R,S] = (U° R) ∪ (V° S) (5.11)	
5	Z : c→D [R,S] Z = [R Z,S Z]	

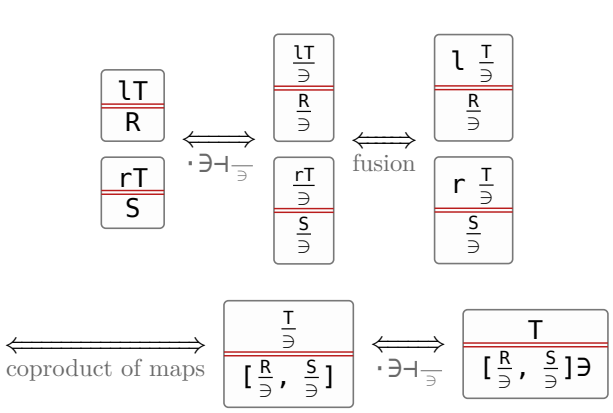
(5.3b)

	the statement R : a_1→b_1, S : a_2→b_2	**picture**
1	R + S ≜ [R l,S r] (5.10)	
2	l l° = 𝟙 , r r° = 𝟙 Ex 5.12	
3	l r° = 𝟘 , r l° = 𝟘 Ex 5.12	
4	(l° l) ∪ (r° r) = 𝟙 Ex 5.12	

§5.3.1. [R,S] = [$\frac{R}{∋}$,$\frac{S}{∋}$] ∋ — a case of relations is the case of their transposes, then ∋

The universal-property row is not free: l,r were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because $\frac{\ }{∋}$ sends an arrow A→C to a map A→PC reversibly, so the map coproduct can be applied underneath it. For any T : A+B→C,

(5.3.1a)

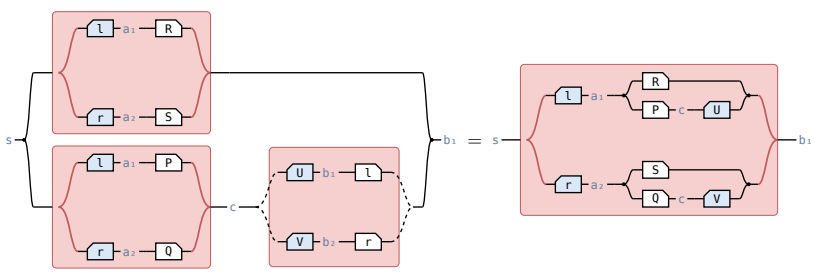


(5.3.1b)

$$\begin{tikzcd}
 & & A \arrow[dl, "\mathtt{l}"'] \arrow[dr, "\frac{R}{\ni}"] \arrow[drr, bend left, "R"] & & \\
A+B \arrow[rrr, dashed, "{[\frac{R}{\ni},\frac{S}{\ni}]}"] & & & PC \arrow[r, "\ni"] & C \\
 & & B \arrow[ul, "\mathtt{r}"] \arrow[ur, "\frac{S}{\ni}"'] \arrow[urr, bend right, "S"'] & &
\end{tikzcd}$$

Nothing here holds only up to ⊑: every triangle commutes on the nose, which is the difference from the fork above. The border spells `[R,S]=[`$\frac{R}{\ni}$`, `$\frac{S}{\ni}$`]∋`, and pushing ∋ into the union that `[·,·]` on maps already is turns that back into the definition, `(l°`$\frac{R}{\ni}$` ∪r°`$\frac{S}{\ni}$`)∋=l°R ∪ r°S`.

(5.3.1c)

	the law	picture
1	prove (5.11), and say why duality does not carry it over from the product law	
2	`(R + S) ∩ ([P,Q] [U,V]°) = (R ∩ (P U°)) + (S ∩ (Q V°))` `[P,Q][U,V]°` is a full 2×2 of composites; `R+S` is diagonal, so the meet cuts the two off-diagonal branches	

§5.4. The power relator P(R)

(5.4a)

For R : A→B,

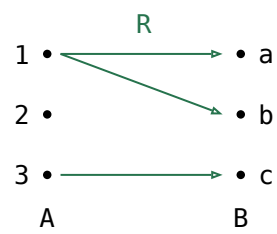
```
          P(R) ≜ (∈ \ (R ∈)) ∩ ((∋ R) / ∋)
E(R)≜ ∋R/∋ = ((∋R)/∋)∩(∋/(∋R))°
xs P(R) ys⟺(∀a∈xs. ∃b∈ys. a R b)∧(∀b∈ys. ∃a∈xs. a R b)
```

Every element of xs is related by R to some element of ys, and conversely.

(5.4b)

		in Rel	in words	im(xs)	partner
1	(∋R)/∋	∀y∈ys. ∃x∈xs. x R y	∀y. some xs R y	ys⊆im(xs)	every y has a partner
2	(∋/(∋R))°	∀y. (∃x∈xs. x R y)→y∈ys	every y with some xs R y is in ys	ys⊇im(xs)	ys leaves out no partner
3	E(R)	ys={y|∃x∈xs. x R y}	ys = every y with some xs R y	ys=im(xs)	every y has a partner, and none is left out
4	((∋R°)/∋)°	∀x∈xs. ∃y∈ys. x R y	∀x. x R some ys	—	every x has a partner
5	P(R)	∀y∈ys. ∃x∈xs. x R y and ∀x∈xs. ∃y∈ys. x R y	∀y. some xs R y and ∀x. x R some ys	—	every x and every y has a partner

(5.4c)



xs	(∋R)/∋	(∋/(∋R))°	E(R)	((∋R°)/∋)°	P(R)
1 ∅	∅	all 8 subsets of abc	∅	all 8 subsets of abc	∅
2 {1}	∅, a, b, ab	ab, abc	ab	a, b, ab, ac, bc, abc	a, b, ab
3 {2}	∅	all 8 subsets of abc	∅	none	none
4 {3}	∅, c	c, ac, bc, abc	c	c, ac, bc, abc	c
5 {1,2}	∅, a, b, ab	ab, abc	ab	none	none
6 {1,3}	all 8 subsets of abc	abc	abc	ac, bc, abc	ac, bc, abc
7 {2,3}	∅, c	c, ac, bc, abc	c	none	none
8 {1,2,3}	all 8 subsets of abc	abc	abc	none	none

(5.4d)

	law	the reading
1	X ⊑ P(R) ⟺ X ∋ ⊑ ∋ R ∧ X° ∋ ⊑ ∋ R°	One containment, and the same one at R° — which is the definition read off the two divisions. Hence P(R°)=P(R)°, and R⊑S⟹P(R)⊑P(S) .
2	P(𝟙) = 𝟙	The straightness axiom verbatim: extensionality **is** P's unit law.
3	Map(f) ⟹ P(f) = E(f)	In Rel, xs P(f) ys⟺ys={f(a)|a∈xs}. The half at f° says every a∈xs has its f(a) on ys; f has just the one image per a, so that already says ys contains everything xs reaches, which is the fraction's second half. For a map the two definitions coincide.
4	P(R S) = P(R) P(S)	⊒ is the division cancellation laws. ⊑ is the one law in this section that is not a calculation: it needs a tabulation of P(RS).

§5.4.1. Type relator

(5.4.1a)

Let F be a binary relator with initial type (α,T), so T is a type functor. F(R,S) is its action on a pair, and F(X) abbreviates F(𝟙,X), the F of the reduce section. For every object A the initial algebra is α : F(A,TA)→TA, among the maps. T acts on an arrow R : A→B by

```
T(R) = ⦇F(R,𝟙) α⦈
```

(5.4.1b)

	name	law	what it says
1	the defining equation	T(R) = ⦇F(R,𝟙) α⦈	Rebuild the structure with α, applying R to the parameter on the way.
2	type relator	T(R)° = T(R°)	A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.

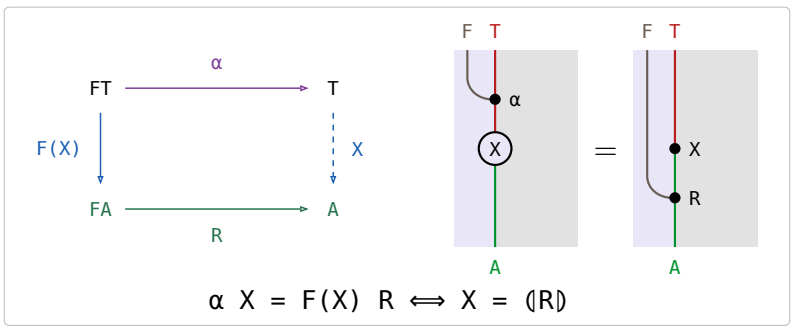
§5.5. Relational catamorphisms

(5.5a)

let F be a relator and has **initial algebra** α : F(T)→T in the subcategory of functions. α is also initial in the allegory:

§5.5.1. The defining equation

(5.5.1a)



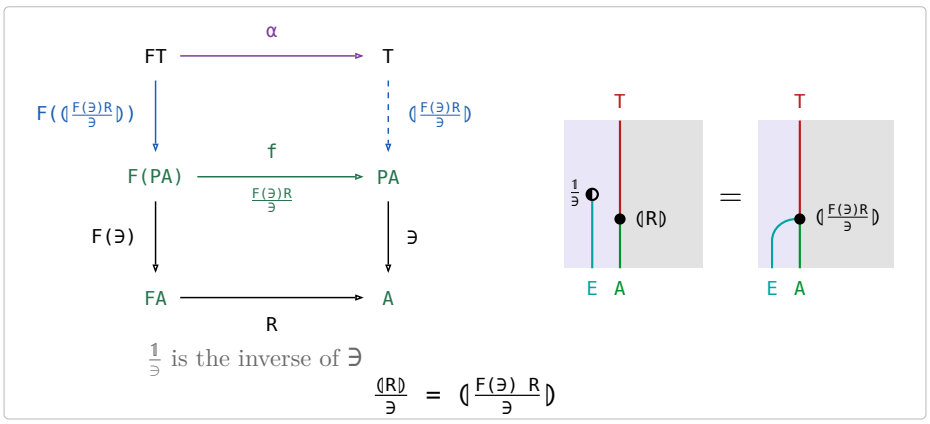
An F-algebra is a **weakened** natural transformation. A transformation F⇒𝟙 on $\mathcal{A}$ would need a component FX→X at every object and a commuting square at every arrow, but F-Algebra only need it works on T and A.

(5.5.1b)

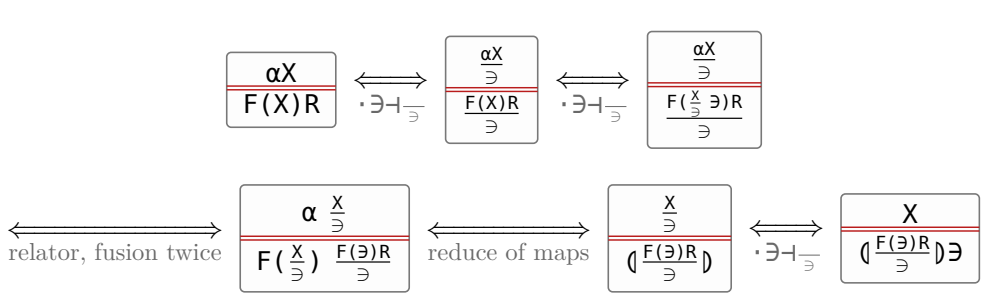
⦇[c,f]⦈=reduce(c,f) catamorphism: arrows of the form ⦇f⦈ are called catamorphisms, and these two examples reveal the notion to be a familiar idea in abstract clothing		
part	**Nat**	**[A]**
datatype	Nat::=zero|succ Nat	[A]::=nil|cons(A,[A])
base functor F	F(X)=1+X	F(X)=1+A×X
initial algebra α	α=[zero,succ] : 1+Nat→Nat	α=[nil,cons] : 1+A×[A]→[A]
the fold, pointwise	⦇[c,f]⦈(zero)=c ⦇[c,f]⦈(succ(n))=f(⦇[c,f]⦈(n))	⦇[c,f]⦈(nil)=c ⦇[c,f]⦈(cons(a,x))=f(a,⦇[c,f]⦈(x))

§5.5.2. ⦇R⦈ = ⦇$\frac{F(\ni)\ R}{\ni}$⦈ ∋ — folding with R is a fold of maps into sets, then ∋

(5.5.2a)



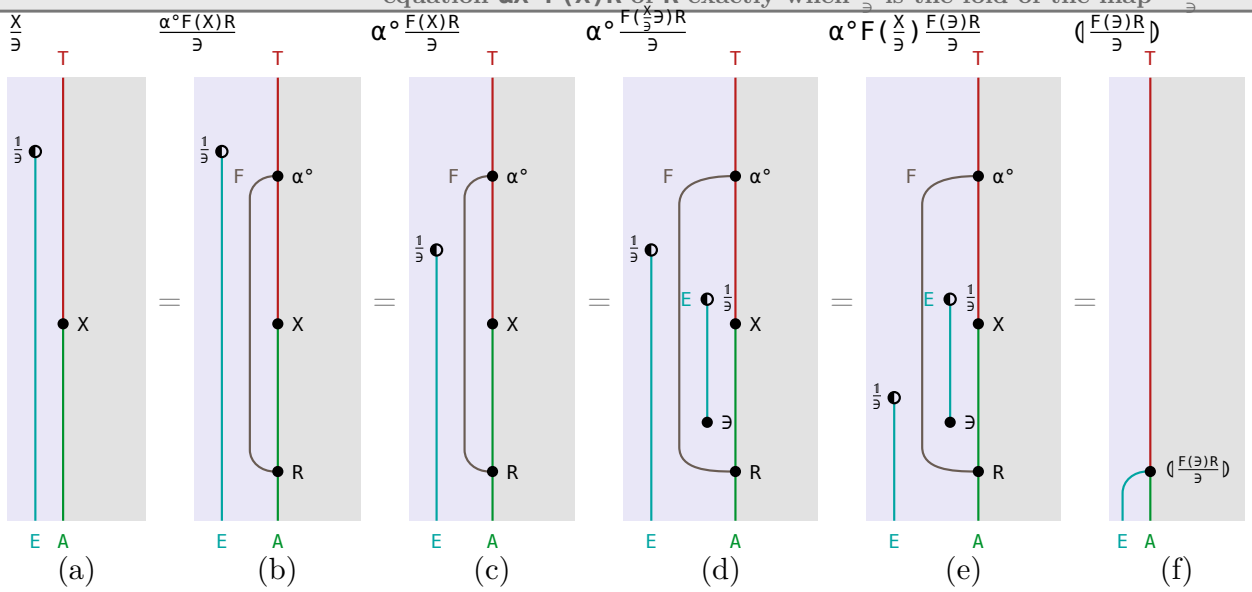
(5.12)



the last two rows at X:=⦇R⦈:

$$\alpha\ X = F(X)\ R \iff \frac{X}{\ni} = (\!|\frac{F(\ni)\ R}{\ni}|\!) \qquad (5.5.2b)$$

in a tabular allegory, with F a relator and $\alpha : F(T) \to T$ its initial algebra: a relation X out of T satisfies the fold equation $\alpha X = F(X)R$ of R exactly when $\frac{X}{\ni}$ is the fold of the map $\frac{F(\ni)R}{\ni}$



(a)	$\frac{X}{\ni}$	α iso; $\alpha X = F(X)R$
(b)	$= \frac{\alpha^\circ\ F(X)\ R}{\ni}$	α° is a map because α is an iso, so it leaves Λ: $\frac{\alpha^\circ F(X)R}{\ni} = \alpha^\circ \frac{F(X)R}{\ni}$ — $Map(f)$ $\Longrightarrow \frac{f\ R}{\ni} = f\ \frac{R}{\ni}$
(c)	$= \alpha^\circ\ \frac{F(X)\ R}{\ni}$	$X = \frac{X}{\ni}\ni$ — $\frac{R}{\ni}\ \ni = R$
(d)	$= \alpha^\circ\ \frac{F(\frac{X}{\ni}\ \ni)\ R}{\ni}$	$F(\frac{X}{\ni}\ni) = F(\frac{X}{\ni})F(\ni)$ as F is a functor, and $F(\frac{X}{\ni})$ is a map (a relator sends maps to maps), so it leaves Λ — $\triangleq$ relator, $Map(f)$ $\Longrightarrow \frac{f\ R}{\ni} = f\ \frac{R}{\ni}$
(e)	$= \alpha^\circ\ F(\frac{X}{\ni})\ \frac{F(\ni)\ R}{\ni}$	fold uniqueness, $\frac{F(\ni)R}{\ni}$ being a map — $(h : f \to g) \triangleq f\ h = F(h)\ g$, $Map(f)$ $\Longrightarrow \alpha\ (\!|f|\!) = F((\!|f|\!))\ f$
(f)	$= (\!|\frac{F(\ni)\ R}{\ni}|\!)$	

§5.6. Combinatorial functions

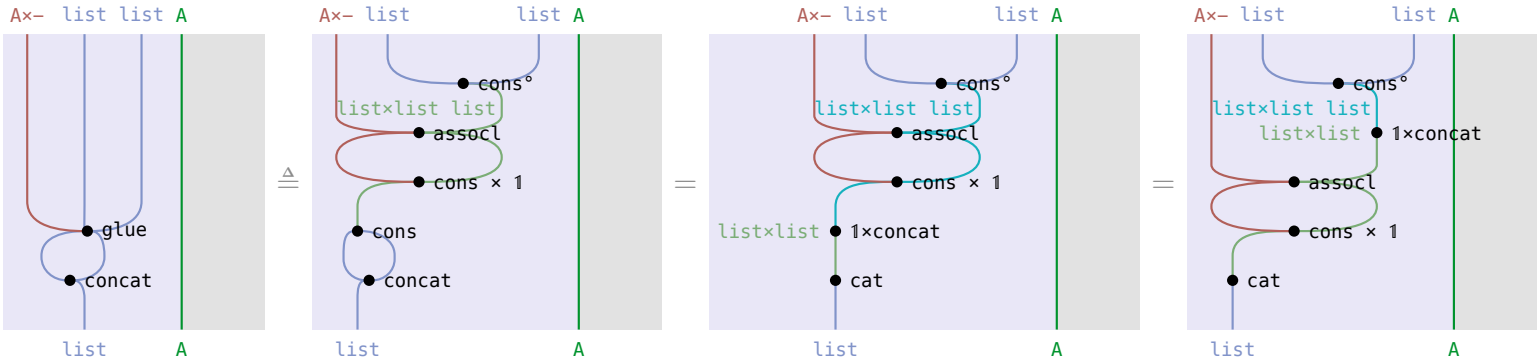
(5.6a)

	definition	type	note
1	`[A]::=nil|cons(A,[A])` `list(A) = [A]`	$\mathcal{A}\to\mathcal{A}$	The list type, under the short name it keeps.
2	`list(R) = ⦇[nil,(R × 𝟙) cons]⦈`	`[A]→[B]`	The relator's action on `R : A→B`: one `R` per element, the shape untouched.
3	`subseq = ⦇[nil,cons ∪ π₂]⦈`	`[A]→[A]`	`xs subseq ys`: `ys` is `xs` with elements dropped — `cons` keeps the head, `π₂` drops it.
4	`prefix = ⦇[nil,⊸ nil ∪ cons]⦈` `prefix = cat° π₁ =init*`	`[A]→[A]`	`ys` is an initial segment of `xs`; the first `nil` is where it stops early. `init≜snoc° π₁`.
5	`suffix = cat° π₂ =tail*`	`[A]→[A]`	The dual, `tail≜cons° π₂`; as a reduce it needs snoc-lists.
6	`partition = concat°`	`[A]→[[A]]`	This `cat` is restricted to `[A]⁺×[A]→[A]`, so `ys` is a list of non-empty segments of `xs`.
7	`concat = ⦇[nil,cat]⦈`	`[[A]]→[A]`	Joins the segments back up, which is why its converse splits a list.
8	`inits`	`[A]→[[A]]`	Implements $\frac{\texttt{prefix}}{\ni}$, listing the prefixes by increasing length.
9	`tails`	`[A]→[[A]]`	Implements $\frac{\texttt{suffix}}{\ni}$ by decreasing length — the opposite order.
10	`cpp`, `cpl`, `cpr`	`[A]×[B]→[A×B]` `[A]×B→[A×B]` `A×[B]→[A×B]`	Every pair `(a,b)` with `a` from the first list and `b` from the second; `cpl`, `cpr` keep one side a single element.
11	`cpp setify = (setify × setify)` $\frac{\ni\times\ni}{\ni}$ `cpl setify = (setify × 𝟙)` $\frac{\ni\times\mathbb{1}}{\ni}$ `cpr setify = (𝟙 × setify)` $\frac{\mathbb{1}\times\ni}{\ni}$	`[A]×[B]→P(A×B)`	Forgetting the order with `setify`, `cpp` is the transpose of `∋×∋`.
12	`cp(F)` $\triangleq \frac{\texttt{F}(\ni)}{\ni}$	`F(PA)→PFA`	`cp(F)`, the transpose of a relator's action on `∋`; `cpp` implements it at `F(A)=A×A`.
13	`cp(list) = ⦇`$\frac{\texttt{[nil,(∋ × ∋) cons]}}{\ni}$`⦈` $\frac{\texttt{[nil,(∋ × ∋) cons]}}{\ni}$ `= [nil` $\frac{\mathbb{1}}{\ni}$`,` $\frac{\ni\times\ni}{\ni}$ `E(cons)]`	`list(PA)→P(list(A))`	`cp(list)`: (5.12) `⦇R⦈ = ⦇`$\frac{\texttt{F(∋) R}}{\ni}$`⦈ ∋` at `list(∋)=⦇[nil, (∋×𝟙)cons]⦈`, then the algebra expanded.
14	`cplist = ⦇[nil wrap,cpp list(cons)]⦈`	`[[A]]→[[A]]`	`cp(list)` with every set a list: `cpp` in place of $\frac{\ni\times\ni}{\ni}$.

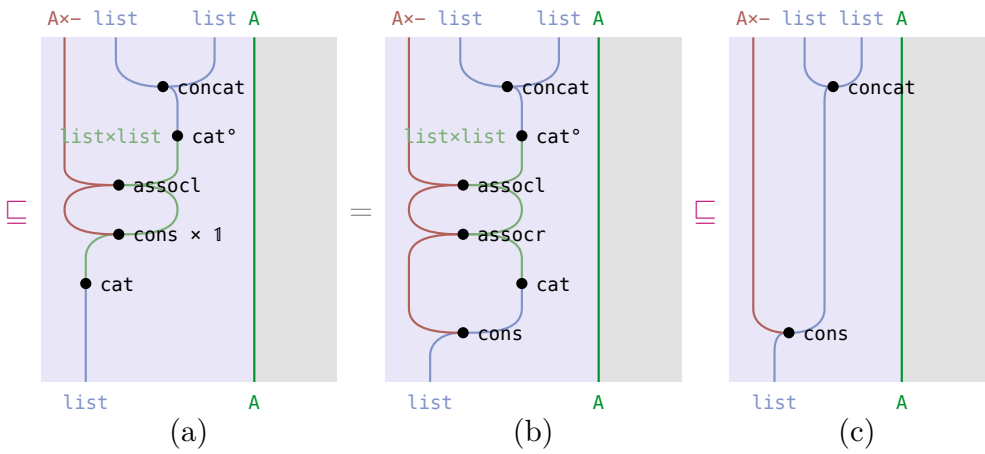
glue concat ⊑ (𝟙 × concat) cons (5.6b)

putting **a** on the front of the first segment and flattening (**glue concat**) gives at most flattening the segments and putting **a** on the front of the result (**(𝟙×concat) cons**)

the starting composite



(a)	glue concat	definition of glue
(b)	≜ (𝟙 × cons°) assocl cons × 𝟙 cons concat	since concat=⦇[nil,cat]⦈
(c)	= (𝟙 × cons°) assocl cons × 𝟙 (𝟙 × concat) cat	naturality of assocl
(d)	= (𝟙 × (cons° (𝟙 × concat))) assocl cons × 𝟙 cat	since concat=⦇[nil,cat]⦈



(a)	⊑ (𝟙 × concat) (𝟙 × cat°) assocl cons × 𝟙 cat	since (cons×𝟙) cat=assocr (𝟙×cat) cons
(b)	= (𝟙 × concat) (𝟙 × cat°) assocl assocr (𝟙 × cat) cons	since assocl assocr=𝟙 and cat° cat⊑𝟙
(c)	⊑ (𝟙 × concat) cons	

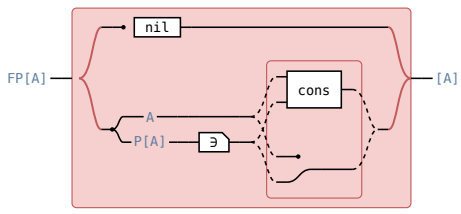
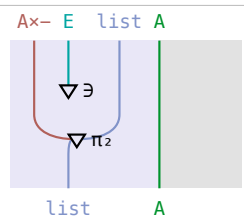
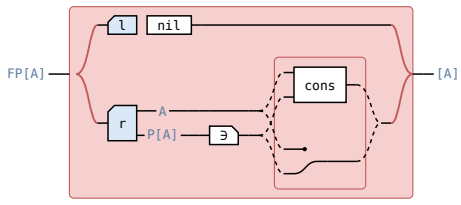
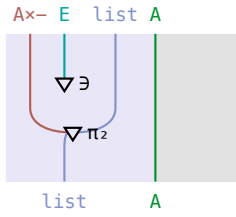
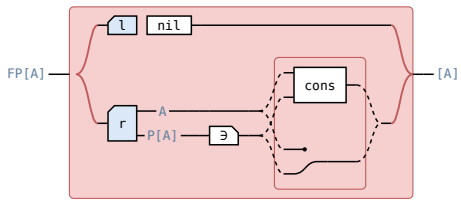
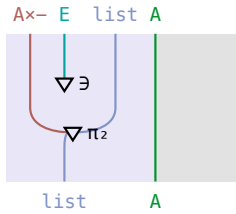
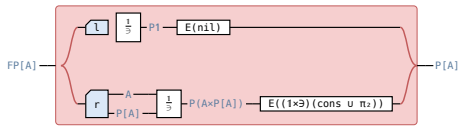
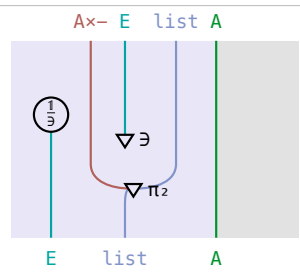
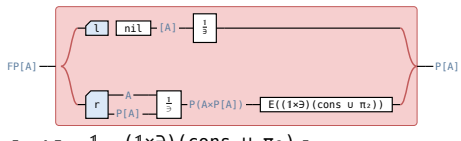
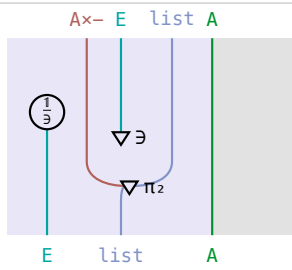
§5.6.1. $\frac{\text{subseq}}{\ni}$ = ⦇[nil $\frac{\mathbb{1}}{\ni}$,⟨$\frac{\mathbb{1}\times\ni}{\ni}$ E(cons),π₂⟩ cup]⦈ — the set of subsequences is a fold of maps into sets

(5.6.1a)

cup≜ $\frac{\pi_1\ni \cup \pi_2\ni}{\ni}$: PA×PA→PA, so $\frac{R \cup S}{\ni}$ =⟨$\frac{R}{\ni}$, $\frac{S}{\ni}$⟩ cup.

$$\frac{F(\ni)\ [nil,cons \cup \pi_2]}{\ni} = [nil\ \frac{\mathbb{1}}{\ni}, \frac{(\mathbb{1} \times \ni)\ (cons \cup \pi_2)}{\ni}] \tag{5.6.1b}$$

the set of lists the algebra builds is, from nothing, just `nil`, and from a head and a set of tails, every tail in the set with the head put on or left off — (5.12) $(\!|R|\!) = (\!|\frac{F(\ni)\ R}{\ni}|\!)\ \ni$ at `subseq=(|[nil,cons ∪ π₂]|)`, `subseq = (|[nil,cons ∪ π₂]|)` .

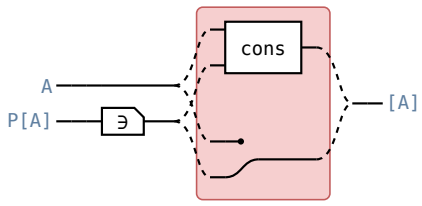
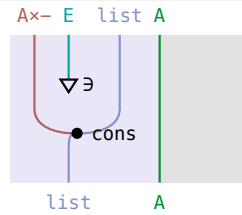
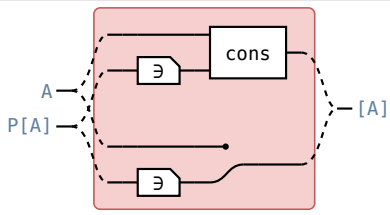
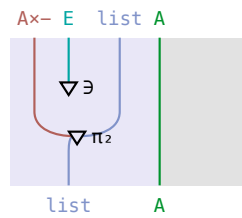
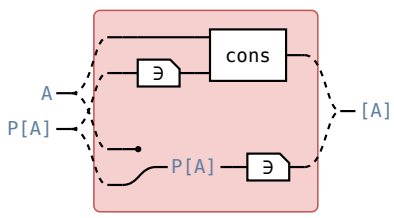
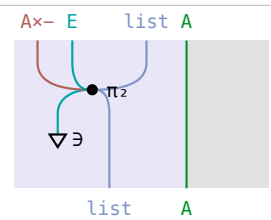
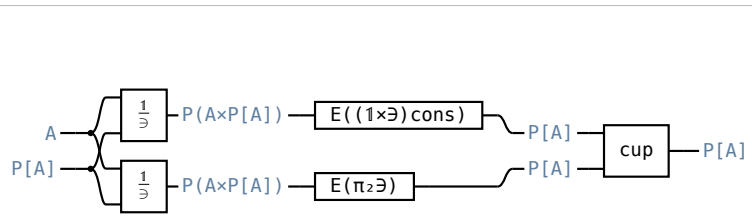
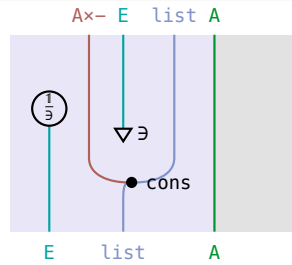
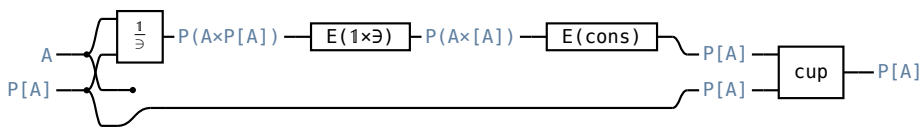
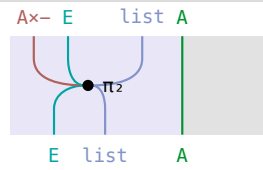
circuit — the fork is `F([A])=1+A×[A]`: `nil` above, the pair below	**Hinze–Marsden**
 $\frac{F(\ni)[nil,cons \cup \pi_2]}{\ni}$	 the π_2 operand of `cons ∪ π₂` under the $\mathbb{1}\times\ni$ summand of `F(∋)`, i.e. $(\mathbb{1}\times\ni)\pi_2$
=  $\frac{(\mathbb{1}+\mathbb{1}\times\ni)[nil,cons \cup \pi_2]}{\ni}$ `F(X)=1+A×X` — `list(A) = [A]`	 the same operand under $\mathbb{1}+\mathbb{1}\times\ni$, whose $\mathbb{1}\times\ni$ summand it sits in
=  $\frac{[nil,(\mathbb{1}\times\ni)(cons \cup \pi_2)]}{\ni}$ $R+S \triangleq [Rl,Sr]$, $l[R,S]=R$, $r[R,S]=S$ — $[R,S] \triangleq (l^\circ\ R) \cup (r^\circ\ S)$, $l\ [R,S] = R$, $r\ [R,S] = S$, $l\ T = R \wedge r\ T = S$ $\Longrightarrow T = [R,S]$	 the π_2 operand of the second arm $(\mathbb{1}\times\ni)(cons \cup \pi_2)$
=  $[\frac{nil}{\ni}, \frac{(\mathbb{1}\times\ni)(cons \cup \pi_2)}{\ni}]$ $[R,S] = [\frac{R}{\ni}, \frac{S}{\ni}]\ \ni$ at `T:=[nil, (1×∋)(cons ∪ π₂)]`	 the π_2 operand under its $\frac{\mathbb{1}}{\ni}$, the arm $(\mathbb{1} \times \ni)\ \pi_2 = \pi_2\ \ni$'s square rewrites, $(\mathbb{1}\times\ni)\pi_2=\pi_2\ni$
=  $[nil\ \frac{\mathbb{1}}{\ni}, \frac{(\mathbb{1}\times\ni)(cons \cup \pi_2)}{\ni}]$ $Map(f) \wedge \mathbb{1}\Box = \ni\Box$ $\Longrightarrow \frac{f}{\ni} = f\ \frac{\mathbb{1}}{\ni}$, $\frac{f}{\ni} = f\ \frac{\mathbb{1}}{\ni}$ for `f` a map, at `f:=nil`	 the same operand; the two rows differ only in the `nil` arm

(5.6.1c)

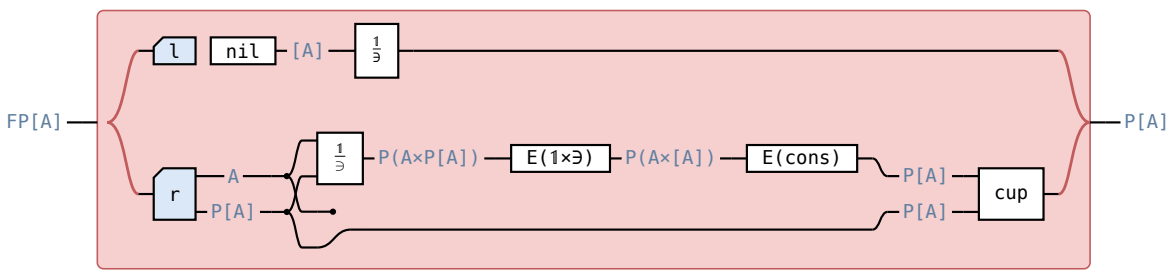
$$\begin{tikzcd}
A\times P[A] \arrow[r, "\pi_2"] \arrow[d, "\mathbb{1}\times\ni"'] & P[A] \arrow[d, "\ni"] \\
A\times[A] \arrow[r, "\pi_2"'] & {[A]}
\end{tikzcd}$$

$$\frac{(\mathbb{1}\times\ni)\ (\mathtt{cons}\cup\pi_2)}{\ni} = \langle \frac{\mathbb{1}\times\ni}{\ni}\ \mathtt{E(cons)},\pi_2\rangle\ \mathtt{cup} \tag{5.6.1d}$$

power transpose of join: the power transpose of the join of two relations is $\langle \frac{R}{\ni}, \frac{S}{\ni}\rangle$ **cup**, where **cup** is the function that returns the union of two sets

circuit	Hinze–Marsden
 $\frac{(\mathbb{1}\times\ni)(\mathtt{cons}\cup\pi_2)}{\ni}$ $\frac{\mathtt{F}(\ni)\ [\mathtt{nil},\mathtt{cons}\cup\pi_2]}{\ni} = [\mathtt{nil}\ \frac{1}{\ni}, \frac{(\mathbb{1}\times\ni)\ (\mathtt{cons}\cup\pi_2)}{\ni}]$'s second branch	 the cons operand of $\mathtt{cons}\cup\pi_2$
=  $\frac{(\mathbb{1}\times\ni)\mathtt{cons}\cup(\mathbb{1}\times\ni)\pi_2}{\ni}$ $\mathtt{T}(\mathtt{X}_1\cup\mathtt{X}_2)=\mathtt{TX}_1\cup\mathtt{TX}_2$ — $(\mathtt{S}\cup\mathtt{T})\ \mathtt{R} = (\mathtt{S}\ \mathtt{R})\cup(\mathtt{T}\ \mathtt{R})$, $\mathtt{R}\ (\mathtt{S}\cup\mathtt{T}) = (\mathtt{R}\ \mathtt{S})\cup(\mathtt{R}\ \mathtt{T})$, $\mathtt{R}\cap(\mathtt{S}\cup\mathtt{T}) = \mathtt{R}\cap\mathtt{S}\cup\mathtt{R}\cap\mathtt{T}$, $(\mathtt{R}\cup\mathtt{S})^\circ = \mathtt{S}^\circ\cup\mathtt{R}^\circ$	 the π_2 operand of $(\mathbb{1}\times\ni)\mathtt{cons}\cup(\mathbb{1}\times\ni)\pi_2$
=  $\frac{(\mathbb{1}\times\ni)\mathtt{cons}\cup\pi_2\ni}{\ni}$ $(\mathbb{1}\times\ni)\pi_2=\pi_2\ni$ — (5.2.1a) at π_2, an equality because $\mathbb{1}$ is entire	 the π_2 operand of $(\mathbb{1}\times\ni)\mathtt{cons}\cup\pi_2\ni$
=  $\langle \frac{(\mathbb{1}\times\ni)\mathtt{cons}}{\ni}, \frac{\pi_2\ni}{\ni}\rangle$ **cup** $\frac{\mathtt{R}\cup\mathtt{S}}{\ni} = \langle \frac{\mathtt{R}}{\ni}, \frac{\mathtt{S}}{\ni}\rangle$ **cup** — $\triangleq$ **cup**	 the cons operand under its $\frac{1}{\ni}$
=  $\langle \frac{\mathbb{1}\times\ni}{\ni}\ \mathtt{E(cons)},\pi_2\rangle$ **cup** $\frac{\mathtt{S}}{\ni}\ \mathtt{E(R)} = \frac{\mathtt{S\ R}}{\ni}$, absorption $\frac{\mathtt{S}}{\ni}\ \mathtt{E(R)}=\frac{\mathtt{SR}}{\ni}$ at $\mathtt{S}:=\mathbb{1}\times\ni$, $\mathtt{R}:=\mathtt{cons}$; fusion and $\frac{\ni}{\ni}=\mathbb{1}$ on the π_2 operand	 the π_2 operand, bare π_2

(5.6.1e)



$$[\frac{\mathtt{nil}}{\ni}, \langle \frac{\mathbb{1}\times\ni}{\ni}\ \mathtt{E(cons)},\pi_2\rangle\ \mathtt{cup}]$$

which writes $\mathtt{Pcons}$; $\mathtt{cons}$ is a map, and there $\mathtt{P(cons)=E(cons)}$ — $\mathtt{Map(f)}$

$$\Longrightarrow \mathtt{P(f)} = \mathtt{E(f)}\ .$$

§5.7. Lax natural transformations (LaT)

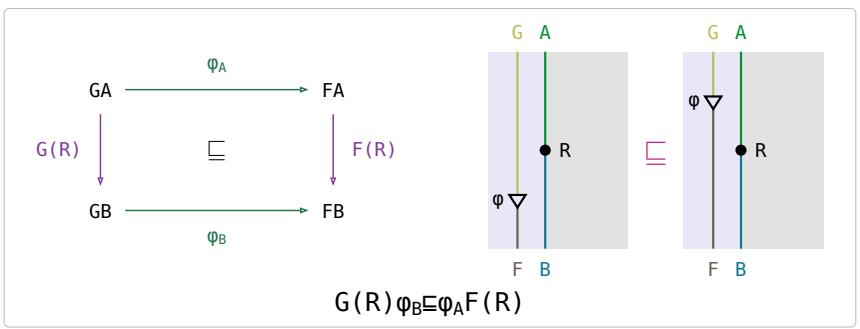
(5.7a)

For relators $G,F : \mathcal{C}\to\mathcal{D}$ and components $\varphi_A : GA\to FA$, φ is **lax at** $R : A\to B$ when $G(R)\varphi_B\sqsubseteq\varphi_A F(R)$ — both sides $GA\to FB$, the left through the component at B, the right through the one at A.

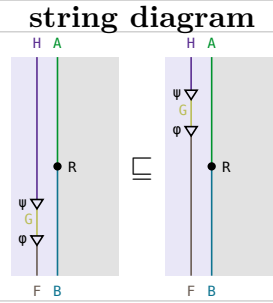
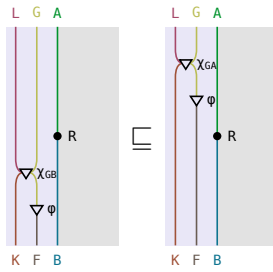
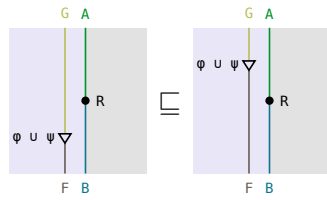
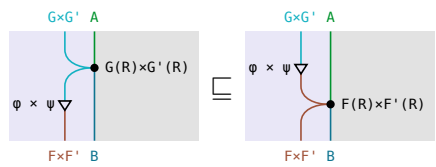
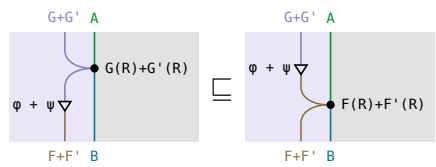
$\varphi : G\Rightarrow F$ is a **lax natural transformation** (LaT) when it is lax at **every** R. (5.13)

Lax at every **map** already gives LaT, and at a map the inequation is an equality $G(f)\varphi=\varphi F(f)$: laxness is about relations only.

(5.7b)



(5.7c)

	closed under	commutative diagram	string diagram
1	composition $\psi\varphi$	$\begin{tikzcd} HA \arrow[r, "\psi_A"] \arrow[d, "H(R)"'] \arrow[dr, phantom, "\sqsubseteq"] & GA \arrow[r, "\varphi_A"] \arrow[d, "G(R)"'] \arrow[dr, phantom, "\sqsubseteq"] & FA \arrow[d, "F(R)"] \\ HB \arrow[r, "\psi_B"'] & GB \arrow[r, "\varphi_B"'] & FB \end{tikzcd}$ $H(R)\psi_B \sqsubseteq \psi_A G(R)$ and $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$ give $H(R)(\psi_B\varphi_B) \sqsubseteq (\psi_A\varphi_A)F(R)$	
2	horizontal composition $\chi\circ\varphi$	$\begin{tikzcd} LGA \arrow[r, "\chi_{GA}K(\varphi_A)"] \arrow[d, "L(G(R))"'] \arrow[dr, phantom, "\sqsubseteq"] & KFA \arrow[d, "K(F(R))"] \\ LGB \arrow[r, "\chi_{GB}K(\varphi_B)"'] & KFB \end{tikzcd}$ $\varphi : G \Rightarrow F$ with $G,F : \mathcal{C}\to\mathcal{D}$ and $\chi : L \Rightarrow K$ with $L,K : \mathcal{D}\to\mathcal{E}$ give $\chi\circ\varphi : L\circ G \Rightarrow K\circ F$, the family $A \mapsto \chi_{GA}K(\varphi_A)$ $L(G(R))\chi_{GB} \sqsubseteq \chi_{GA}K(G(R))$ is χ lax at $G(R)$; then $K(G(R)\varphi_B) \sqsubseteq K(\varphi_A F(R))$ is K applied to φ's own inequation; $\chi := \mathbb{1}_K$ gives $K(\varphi) : K\circ G \Rightarrow K\circ F$, K composed on the outside; $\varphi := \mathbb{1}_G$ gives $\chi G : L\circ G \Rightarrow K\circ G$, G composed on the inside two candidates, $\chi_{GA}K(\varphi_A)$ and $L(\varphi_A)\chi_{FA}$, ordered by $\sqsubseteq$; this row is the first	
3	union $\varphi \cup \psi$	$\begin{tikzcd} GA \arrow[r, "\varphi_A"] \arrow[d, "G(R)"'] \arrow[dr, phantom, "\sqsubseteq"] & FA \arrow[d, "F(R)"] \\ GB \arrow[r, "\varphi_B"'] & FB \end{tikzcd} \cup \begin{tikzcd} GA \arrow[r, "\psi_A"] \arrow[d, "G(R)"'] \arrow[dr, phantom, "\sqsubseteq"] & FA \arrow[d, "F(R)"] \\ GB \arrow[r, "\psi_B"'] & FB \end{tikzcd}$ $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$ and $G(R)\psi_B \sqsubseteq \psi_A F(R)$ give $G(R)(\varphi_B \cup \psi_B) \sqsubseteq (\varphi_A \cup \psi_A)F(R)$	
4	a relator K $K(\varphi)$	$\begin{tikzcd} KGA \arrow[r, "K(G(R))"] \arrow[d, "K(\varphi_A)"'] \arrow[dr, phantom, "\sqsubseteq"] & KGB \arrow[d, "K(\varphi_B)"] \\ KFA \arrow[r, "K(F(R))"'] & KFB \end{tikzcd}$ $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$ gives $K(G(R))K(\varphi_B) \sqsubseteq K(\varphi_A)K(F(R))$	
5	product $\varphi\times\psi$	$\begin{tikzcd} GA\times G'A \arrow[r, "\varphi_A\times\psi_A"] \arrow[d, "G(R)\times G'(R)"'] \arrow[dr, phantom, "\sqsubseteq"] & FA\times F'A \arrow[d, "F(R)\times F'(R)"] \\ GB\times G'B \arrow[r, "\varphi_B\times\psi_B"'] & FB\times F'B \end{tikzcd}$ $\varphi : G \Rightarrow F$ and $\psi : G' \Rightarrow F'$ give $\varphi\times\psi : G\times G' \Rightarrow F\times F'$ $(R\times S)(U\times V) = (RU)\times(SV)$ and monotonicity in both slots — the row above's K applied to the inequation, run on a bifunctor; the fork is the derived case $\langle\varphi,\psi\rangle = \triangleleft(\varphi\times\psi)$, and its ONE cost is the lax copy law $R\triangleleft \sqsubseteq \triangleleft(R\times R)$ — $R \triangleleft \leq \triangleleft (R \otimes R)$, $R \multimap \leq \multimap$	
6	coproduct $\varphi+\psi$	$\begin{tikzcd} GA+G'A \arrow[r, "\varphi_A+\psi_A"] \arrow[d, "G(R)+G'(R)"'] \arrow[dr, phantom, "\sqsubseteq"] & FA+F'A \arrow[d, "F(R)+F'(R)"] \\ GB+G'B \arrow[r, "\varphi_B+\psi_B"'] & FB+F'B \end{tikzcd}$ $\varphi : G \Rightarrow F$ and $\psi : G' \Rightarrow F'$ give $\varphi+\psi : G+G' \Rightarrow F+F'$ $(R+S)(U+V) = (RU)+(SV)$ and monotonicity in both slots; the cofork is the derived case $[\varphi,\psi] = (\varphi+\psi)\triangledown$, and $\triangledown$ costs nothing	
7	meet — **fails** $\varphi\cap\psi$	the step it would need is $(\varphi_A F(R)) \cap (\psi_A F(R)) \sqsubseteq (\varphi\cap\psi)_A F(R)$, the wrong direction of $R\ (S \cap T) \sqsubseteq (R\ S) \cap (R\ T)$ a counterexample: (5.7.1a)	
8	Each row is the general slide $R_A X \sqsubseteq X' R_B$ and $R_B Y \sqsubseteq Y' R_C$ give $R_A(XY) \sqsubseteq (X'Y')R_C$ at $R_A, R_B := G(R), F(R)$ and $X, X' := \varphi_B, \varphi_A$, quantified over R; at $X' = X$ with R_A, R_B endorelations it is the monotonicity reading instead — the $\forall R$ sits outside the law, and is the only difference		

§5.7.1. meet is not closed in LaT

(5.7.1a)

counterexample — φ,ψ : G⇒F lax natural does NOT give φ∩ψ lax natural

$$\begin{tikzcd}
\begin{matrix} A\times A \\ (0,1) \end{matrix} \arrow[r, "\pi_1\cap\pi_2"] \arrow[d, "R\times R"'] \arrow[dr, phantom, "\not\sqsubseteq"] & \begin{matrix} A \\ \varnothing \end{matrix} \arrow[d, "R"] \\
\begin{matrix} B\times B \\ (0,0) \end{matrix} \arrow[r, "\pi_1\cap\pi_2"'] & \begin{matrix} B \\ \{0\} \\ \varnothing \end{matrix}
\end{tikzcd}$$

A=B≜{0,1}, R≜{(0,0),(1,0)}, φ≜π₁∩π₂ : Δ⇒Id

§5.7.2. Two stacked towers

(5.7.2a)

		lower — inside ONE allegory	upper — between allegories
1	0-cell	an object A	an allegory $\mathcal{C}$ — **a whole lower tower**
2	1-cell	a relation R : A→B	a relator F : $\mathcal{C}\to\mathcal{D}$
3	2-cell	R⊑S	a LaT φ : G⇒F ≜ lax
4	order	R⊑S	φ⊑ψ componentwise
5	union	R ∪ S	φ ∪ ψ, **survives** G(R) (φ_B ∪ ψ_B) ⊑ (φ_A ∪ ψ_A) F(R)
6	product	R×S ≜ ×	φ×ψ, **survives**; a TENSOR, not a categorical product; SYMMETRIC ⟨R,S⟩ π₁ = dom(S) R , ⟨R,S⟩ π₂ = dom(R) S (G(R) × G'(R)) (φ_B × ψ_B) ⊑ (φ_A × ψ_A) (F(R) × F'(R))
7	coproduct	R+S	φ+ψ, **survives**; a BIPRODUCT (G(R) + G'(R)) (φ_B + ψ_B) ⊑ (φ_A + ψ_A) (F(R) + F'(R))
8	meet	R∩S	φ∩ψ componentwise, **fails** (5.7.1a)
9	converse	R°	φ°, **fails**, oplax: φ_A°G(R)⊑F(R)φ_B°
10	zero object	z with 𝟙 z=𝟘	the constant relator at such a z, initial and terminal

§5.7.3. R⊑S is a special case of LaT

(5.7.3a)

$$\begin{tikzcd}
GA \arrow[r, "\varphi_A"] \arrow[d, "G(R)"'] \arrow[dr, phantom, "\sqsubseteq"] & FA \arrow[d, "F(R)"] \\
GB \arrow[r, "\varphi_B"'] & FB
\end{tikzcd}
\Longrightarrow
\begin{tikzcd}
A \arrow[r, "\varphi_X"] \arrow[d, "\mathbb{1}"'] \arrow[dr, phantom, "\sqsubseteq"] & B \arrow[d, "\mathbb{1}"] \\
A \arrow[r, "\varphi_Y"'] & B
\end{tikzcd}
\Longrightarrow
\begin{tikzcd}
A \arrow[r, bend left, "\varphi_Y"] \arrow[r, bend right, "\varphi_X"'] \arrow[r, phantom, "\sqsubseteq"] & B
\end{tikzcd}$$

G≜const A, F≜const B — the relators X↦A and X↦B, each sending every arrow to 𝟙 ≜ lax

with $\mathcal{C}$ two objects and one non-identity arrow X→Y, a LaT const A⇒const B is the pair φ_X,φ_Y : A→B with φ_Y⊑φ_X — the datum R⊑S

not every LaT is constant: ∋ : P⇒Id X ⊑ P(R) ⟺ X ∋ ⊑ ∋ R ∧ X° ∋ ⊑ ∋ R° π₂ : (A×−)⇒Id (𝟙 × ∋) π₂ = π₂ ∋

◁ : Id⇒Δ R ◁ ≤ ◁ (R ⊗ R) , R ⊸ ≤ ⊸